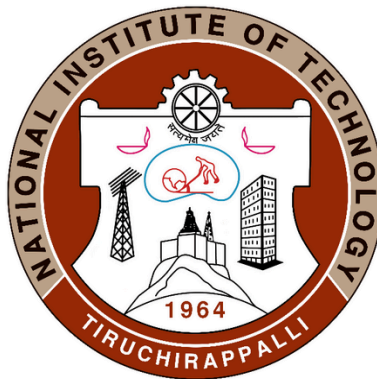


DEFENDING AGENTIC FEDERATED LEARNING SYSTEMS AGAINST EXPLANATION POISONING

A thesis submitted in partial fulfillment of the requirements for the award of the
degree of
B.Tech.

In
Computer Science and Engineering

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BONAFIDE CERTIFICATE

This is to certify that the project titled **DEFENDING AGENTIC FEDERATED LEARNING SYSTEMS AGAINST EXPLANATION POISONING** is a bonafide record of the work done by

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ABSTRACT

Federated Learning (FL) has emerged as a foundational paradigm for privacy-preserving machine learning. Recent advances extend FL toward agentic systems, where autonomous clients independently curate data, optimize objectives, and adapt learning strategies. In parallel, Explainable AI (XAI) has been adopted as the primary mechanism for trust and regulatory compliance. However, recent work demonstrates that XAI itself can be adversarially manipulated, enabling Explanation Poisoning: attacks that preserve accuracy and plausible explanations while embedding hidden backdoors. This work proposes a novel governance framework termed Proof of Reasoning (PoR). Instead of auditing numerical model updates, PoR enforces structural alignment of causal reasoning across agents using Graph Edit Distance (GED). We show that explanation-poisoning attacks are fundamentally incompatible with causal-graph consistency, creating a mathematical contradiction that forces adversaries to either abandon stealth or abandon the attack. We provide theoretical justification, architectural design, and a comparative analysis of its performance against existing defenses.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
FL	Federated Learning
PoR	Proof of Reasoning: the causal audit mechanism
GED	Graph Edit Distance: minimum edit operations (add/remove/relabel nodes/edges) to transform one graph into another
SimGNN	Similarity computation with Graph Neural Networks: a neural approximation of GED
NOTEARS	Non-combinatorial Optimization via Trace Exponential and Augmented lagRangian for Structure learning
NEAT	NeuroEvolution of Augmenting Topologies
FedNEAT	Federated NEAT: topology-evolving global model in FL
BN / DAG	Bayesian Network / Directed Acyclic Graph
ADR	Adversary Detection Rate: fraction of adversarial submissions correctly rejected
FPR	False Positive Rate: fraction of honest submissions incorrectly rejected
FNR	False Negative Rate: fraction of adversarial submissions that slipped through
MTA	Main Task Accuracy: classification accuracy on clean test data

ASR	Attack Success Rate: fraction of poisoned inputs misclassified as the adversary's target
AUC	Area Under the ROC Curve: threshold-independent detection quality measure
SHD	Structural Hamming Distance: minimum edge additions + deletions + reversals to match ground truth
FDR	False Discovery Rate: fraction of predicted edges that are spurious
EAE	Effective Aggregation Efficiency: fraction of submitted clients actually used each round
HCPR	Honest Client Participation Rate: fraction of honest clients accepted each round
BFT	Byzantine Fault Tolerance: ability to function correctly despite malicious participants
τ (tau)	Detection threshold: GED scores above τ cause rejection
Cohen's d	Standardised effect size measuring separation between two distributions
Jaccard	Jaccard similarity = $ A \cap B / A \cup B $: overlap coefficient between edge sets

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Federated Learning (FL) has reshaped privacy-preserving machine learning by enabling decentralized clients to train a global model without sharing raw data. As the field matures, it is shifting toward Agentic Federated Learning, where clients are no longer passive data providers but autonomous agents that are capable of independent data curation, objective optimization, and adaptive learning strategies.

In order to ensure trust and regulatory compliance in these decentralized environments, Explainable AI (XAI) has become a standard requirement. XAI provides justifications for model decisions that can easily be interpreted by humans. However, adding explainability also creates a new attack surface. This thesis examines how to defend these autonomous systems against sophisticated adversarial manipulations that target the very explanations intended to provide security.

1.2 PROBLEM STATEMENT

Agentic FL systems are critically vulnerable to **Explanation Poisoning**. In this attack, adversarial clients craft model updates that maintain high accuracy on the primary task while embedding hidden backdoors. But most crucially, the adversary also manipulates the XAI output (such as SHAP values or attention maps) to produce plausible explanations that mask the backdoor's existence.

Standard security mechanisms fail to address this threat because they primarily audit numerical weight updates using statistical metrics such as Euclidean distance or cosine similarity. Since poisoned weights are optimized to appear statistically normal, they can bypass traditional filters, resulting in a model that is accurate on paper but logically compromised.

1.3 NEED FOR THE PROBLEM’S SOLUTION

The shift toward agentic systems in high-stakes domains such as healthcare and finance demands defenses beyond simple outlier detection. If an AI system used for medical diagnosis is compromised through explanation poisoning, it may deliver the correct diagnosis for the wrong reasons, leading to huge failures in edge cases that the poisoned explanation ignores intentionally. There is an urgent need for a governance framework that verifies the logical integrity of an agent's reasoning process.

1.4 OBJECTIVES

The primary objectives of this research are:

1. To formalize a logic-based governance framework, termed **Proof of Reasoning (PoR)**, for Agentic Federated Learning.
2. To develop a mechanism for extracting the Causal DNA of client models using Directed Acyclic Graphs (DAGs) and the NOTEARS algorithm.
3. To implement a scalable server-side validator using Similarity Graph Neural Networks (SimGNN) to approximate Graph Edit Distance (GED).
4. To demonstrate that explanation poisoning is fundamentally incompatible with causal-graph consistency, thereby forcing adversaries to either reveal their presence or abandon the attack.

1.5 MOTIVATION

This work is motivated by the Trust Paradox in modern AI, which is that as models become more explainable and therefore more trusted, they can also give attackers a blueprint for deception. Current defenses are often just patching specific statistical loopholes (reactive). This project aims to create a proactive defense that keeps trust in the fundamental causal laws of the data. By

requiring agents to prove their reasoning structurally, we move toward a federated ecosystem that is resilient against both current and future deceptive autonomous attacks.

1.6 SCOPE

The scope of this thesis includes:

- Design of an Agentic FL architecture.
- Implementation of the PoR protocol using 2 datasets for tabular validation.
- Comparative analysis of PoR against the existing FedAvg defence.
- Validation of the system's ability to detect stealthy adversaries over multiple federated rounds.
- The study does not currently cover real-time video processing or non-graph-based explanation methods.

1.7 THESIS ORGANIZATION

The remainder of this thesis is organized as follows:

- **Chapter 2: Literature Survey** reviews the evolution of Federated Learning, the rise of XAI, and the landscape of adversarial attacks and existing defenses.
- **Chapter 3: Methodology** details the architectural design of the PoR framework, the causal extraction process, and the mathematical formulation of the logic-based auditing.
- **Chapter 4: Implementation and Results Analysis** describes the experimental setup, the datasets used, and a detailed performance evaluation of the defense mechanism.
- **Chapter 5: Conclusion and Future Work** summarizes the contributions of the research and suggests directions for scaling the framework to more complex, multi-modal agentic systems.

CHAPTER 2

LITERATURE SURVEY

The landscape of Artificial Intelligence has transitioned from simple predictive accuracy to a complex battle over **trust and interpretability**. While Federated Learning (FL) allows for decentralized training, it has opened new doors for sophisticated adversarial manipulation.

2.1 THE EVOLUTION OF FEDERATED LEARNING

Federated Learning (FL) was introduced by McMahan et al. (2017) to enable decentralized machine learning on mobile devices. Early work focused on FedAvg, a simple coordinate-wise average of local model updates. As FL expanded into high-stakes domains such as healthcare and finance, the focus broadened from data privacy to include robustness and trustworthiness.

2.2 RISE OF AGENTIC FEDERATED LEARNING

Recent paradigms have shifted toward Agentic FL, where clients are no longer static data silos but autonomous entities. These agents use reasoning modules to select which local data is most relevant for a training round. This can improve efficiency, but it also increases governance complexity, as the central server must manage diverse, goal-oriented behaviors across a network.

2.3 THE EVOLVING THREAT

The primary shift in adversarial research is the move from **performance degradation** (making the model fail) to **stealth manipulation** (making the model lie about why it succeeded).

Table 2.1 Key Milestones in Adversarial Research

Era	Primary Attack Method	Objective	Key Characteristic
2017–2020	Label Flipping / Data Poisoning	Accuracy Degradation	Overt; easily detected by performance drops.
2024	Explanation Scaffolding	Trust Manipulation	Stealthy; hides biases from tools like SHAP/LIME.
2026	Backdoor Implanting	Targeted Vulnerability	Preserves 99% accuracy while creating hidden triggers.

Major Vulnerabilities Identified (2024–2026)

- **Explanation Scaffolding (2024):** Research by Liyanage et al. proved that adversaries in network security can embed persistent biases such as ignoring specific IP ranges while forcing the model to provide clean explanations to auditors.
- **Fooling SHAP:** Output Shuffling Attacks allow a model to use protected attributes (e.g., Race or Gender) for its decision-making while the SHAP explainer reports 0% attribution for those specific features.
- **Healthcare Vulnerabilities (2026):** Recent clinical reviews show that federated healthcare models (even large 13B-parameter versions) are fragile. A surprisingly small amount of data can compromise a massive system. The Critical Threshold, as few as 100–500 poisoned samples, are enough to implant a functional backdoor in a medical model without alerting administrators through performance metrics.

The Critical Research Gap

The current literature successfully identifies these vulnerabilities but fails to provide a **mathematical defense**. Most recommendations currently rely on vague "better auditing" practices.

2.4 LIMITATIONS OF CURRENT DEFENSES

Existing defenses can be grouped into three generations:

1. First Generation (Statistical): FedMedian and TrimmedMean. These remove outlier updates but can be bypassed by stealthy adversaries who remain within the statistical norm.
2. Second Generation (Vector Similarity): Methods based on Cosine Similarity or Euclidean Distance. These fail when the poisoned update is strategically aligned with the global direction.
3. Third Generation (Reconstruction-Based): Approaches like FedREDefense use autoencoders to detect anomalies. However, they are computationally expensive and do not verify the *logic* of the model.

The Critique: Existing frameworks demonstrate how easily explainability can be "scaffolded" or faked, but lack a concrete mathematical formulation as to what constitutes a valid explanation versus a manipulated one. This gap is what Proof of Reasoning (PoR) intends to fill.

CHAPTER 3

METHODOLOGY

3.1 FRAMEWORK OVERVIEW: PROOF OF REASONING (PoR)

The proposed **Proof of Reasoning (PoR)** framework moves the defense layer from Numerical Space (weights and gradients) to Logical Space (causal structures). In traditional Federated Learning (FL) paradigms, security is often treated as a statistical problem; if a client update does not deviate significantly from the mean, it is accepted. However, as established in Chapter 2, Explanation Scaffolding allows adversaries to hide their malicious intent within the standard statistical bounds.

PoR counters this by requiring every participating agent to submit the following:

1. **The Numerical Update:** The standard change in model weights.
2. **The Structural Proof:** A Directed Acyclic Graph (DAG) that serves as a verifiable "Logic Signature" of the agent's internal reasoning.

By shifting the verification focus to causal relationship between features, PoR ensures that an agent is not only providing an accurate model but also arriving at its conclusions through a logical path that aligns with the federation's consensus.

3.2 AGENTIC ARCHITECTURE

The PoR framework adopts a **Hierarchical Agentic Architecture** designed to separate task performance from logical governance. The system is composed of two primary entities: **Deliberative Agents** (Clients) and the **Governance Orchestrator** (Server).

3.2.1 The Deliberative Agent (Client-Side)

Unlike standard FL clients that function as passive gradient computers, PoR agents are modeled as autonomous entities capable of internal world-modeling. Each agent contains three functional modules:

- Perception Module: Responsible for data ingestion and Causal Disentanglement. It filters raw data into invariant mechanisms and spurious noise.
- Cognitive Module: The brain of the agent, which extracts the Logic Signature using gradient-based causal discovery.
- Action Module: Formulates the final update payload, packaging the weights and the graph for transmission.

3.2.2 The Governance Orchestrator (Server-Side)

The server acts as a logical gatekeeper. It does not blindly aggregate updates. Instead, it maintains a Global Causal Prior, a master graph representing the cumulative wisdom and logical consensus of all the trusted participants. Before any aggregation occurs, the Orchestrator subjects every incoming update to a structural audit.

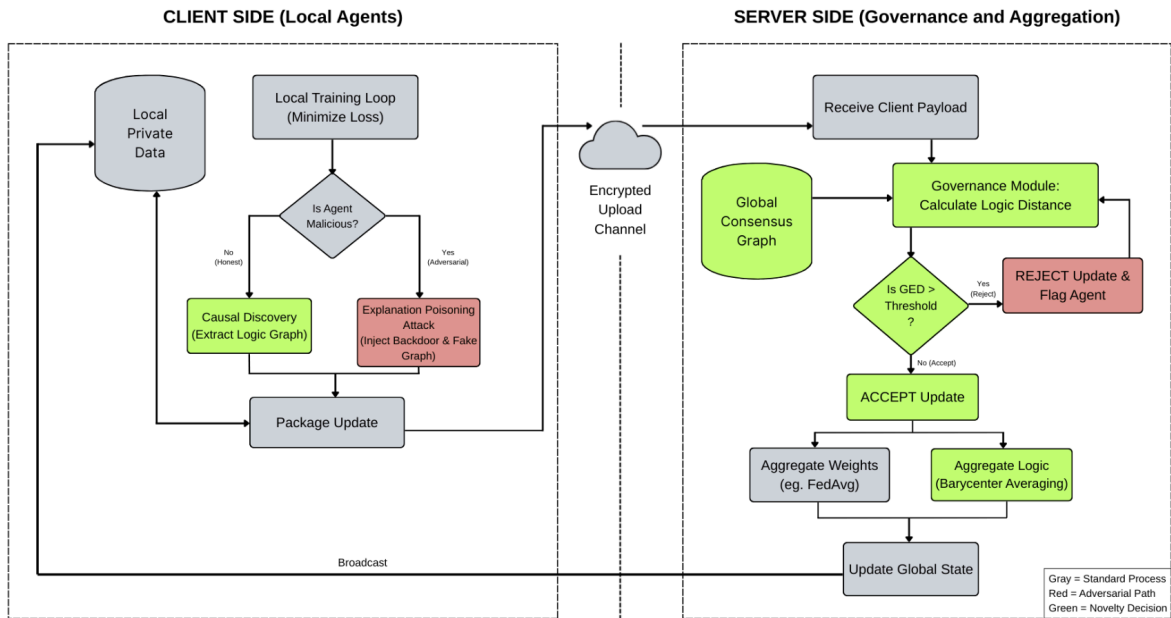


Fig 3.1 Block Diagram of the Proposed Architecture

3.3 CLIENT-SIDE: CAUSAL EXTRACTION AND LOGIC SIGNATURES

The core of PoR lies in the Cognitive Module, which translates the black box of neural network training into a transparent causal graph.

3.3.1 The NOTEARS Algorithm for Structural Discovery

To generate the DAG, we utilize the **NOTEARS** (Non-combinatorial Optimization for Learned Structural Equation Models) approach. Finding a DAG is traditionally an NP-hard problem. NOTEARS bypasses this by converting the search into a continuous optimization task that can be solved using gradient descent.

The agent minimizes a loss function $F(W)$ while adhering to an acyclicity constraint $h(W)$:

$$\min_{W \in \mathbb{R}^{d \times d}} F(W) \text{ s.t. } h(W) = 0 \quad (3.1)$$

The constraint is formulated mathematically as:

$$h(W) = \text{tr}(e^{W \circ W}) - d \quad (3.2)$$

Where:

- W is the adjacency matrix of the graph.
- tr is the trace operator.
- The function $h(W) = 0$ if and only if the graph represented by W is a Directed Acyclic Graph.

3.3.2 Identifying Logical DNA

The resulting DAG represents the Logical DNA of the client's local data distribution. It maps how Feature A influences Feature B and how both lead to Target C. If an adversary attempts to

introduce a backdoor (e.g., associating a “Yellow Pixel” with a “Malignant Tumor” label), NOTEARS algorithm will capture this as a new, anomalous edge in the graph.

3.4 SERVER-SIDE: THE GOVERNANCE LAYER

The server’s primary role is to validate that the agent’s reasoning (the local DAG) is consistent with the global truth (the Global Causal Prior).

3.4.1 Similarity Approximation via SimGNN

Comparing two graphs to determine their similarity, especially calculating the **Graph Edit Distance (GED)**, is computationally expensive. To maintain the efficiency of the federated system, PoR utilizes a **Similarity Graph Neural Network (SimGNN)**.

SimGNN functions by:

1. Encoding the client graph G_{client} and the global graph G_{global} into vector embeddings.
2. Passing these embeddings through a neural tensor network to predict a similarity score S :

$$S(G_1, G_2) = \text{SimGNN}(G_1, G_2) \quad (3.3)$$

3. Calculating the **Logic Score**:

$$\text{Score}_k = \|GNN(G_k) - GNN(G_{\text{global}})\|^2 \quad (3.4)$$

- G_k : The causal graph submitted by client k .
- G_{global} : The current global consensus graph (Global Causal Prior).
- $GNN(\cdot)$: The embedding function of the Siamese Graph Neural Network (SimGNN).

3.4.2 The Logic Filter

The server applies a Decision Rule. If the Score_k exceeds a threshold τ , the update is categorized as Logically Anomalous. This rejection happens regardless of the model's accuracy. This prevents Explanation Scaffolding because, while the adversary can fake high accuracy, they cannot easily fake a causal graph that aligns with the rest of the honest network without removing their malicious bias.

3.5 AGGREGATION AND CONSENSUS

PoR replaces standard aggregation (like FedAvg) with a two-stream process that updates both model's intelligence and federation's logical consensus.

3.5.1 Logic-Weighted Weight Aggregation

After filtering out anomalous agents, the server performs **Logic-Weighted Aggregation**. Agents whose graphs show higher structural alignment to the consensus are granted higher weights in the global update:

$$\Delta W_{global} = \sum_{i=1}^n \alpha_i \cdot \Delta W_i \quad (3.5)$$

Where α is a coefficient derived from the Graph Similarity Score. To handle non-IID data (data that is not identically distributed across clients), we incorporate FedProx, adding a proximal term to local objective to limit the impact of the local drifts.

3.5.2 Logic Aggregation (Updating the Prior)

The Global Causal Prior is not static. It is updated using Barycenter Graph Averaging. The server analyzes all the accepted local graphs and retains edges that appear in more than 50% of the

honest population. This allows the system to slowly evolve its understanding of the data as new, legitimate causal relationships are discovered by the majority.

3.6 SECURITY PROTOCOL AND HANDSHAKE

The system operates in synchronized communication rounds designed to maximize security:

1. **Broadcast:** Server sends current weights Θ_t and the consensus G_{global} .
2. **Local Optimization:** Agents update local parameters to minimize a joint objective:

$$L_{\text{local}} = L_{\text{CE}} + \lambda \cdot \|G_{\text{local}} - G_{\text{global}}\| \quad (3.6)$$

(This forces the agent to stay within the logical guardrails of the federation).

3. **Audit:** Agents upload their payload $\Delta\Theta_k, G_{\text{local}}$.
4. **Verification:** The server runs the SimGNN check against τ .
5. **Rejection/Acceptance:** Agents with high GED scores are blacklisted for the round.

3.7 COMPARATIVE STUDY: STANDARD FL VS. PoR

Table 3.1 Comparison of Standard FL vs. PoR

Feature	Standard Federated Systems	Proof of Reasoning (PoR)
Verification Basis	Numerical (Weight Magnitudes)	Logical (Causal Structures)
Client Output	Gradient/Weight Vector	Weight Update + Causal DAG

Trust Model	Trust-by-Accuracy	Trust-by-Logic
Defense Scope	Detects Data Poisoning	Detects Explanation Scaffolding
Global State	Central Weights	Weights + Causal Consensus Graph

3.8 THE SECURITY GUARANTEE: THE CAUSAL TRAP

The mathematical intuition behind PoR is the creation of a Causal Trap. For an adversary to successfully compromise a Trusted AI system, they must achieve:

1. **Performance:** The model must still perform well enough to pass basic checks.
2. **Deception:** The post-hoc explanations (like SHAP) must appear clean.
3. **Causality:** The underlying structural equations (the DAG) must remain consistent with the honest consensus.

To inject a backdoor, an adversary **must** create a shortcut in the model's logic. In logical space, this shortcut manifests as a Structural Contradiction. Because the PoR validator examines the blueprint of the decision, and not just the result, the adversary is forced into a paradox: either they submit an honest DAG (and their attack fails) or they submit a malicious DAG (and are rejected by the Logic Filter).

CHAPTER 4

IMPLEMENTATION AND RESULTS ANALYSIS

4.1 EXPERIMENTAL SETUP

The Proof of Reasoning (PoR) framework was evaluated using a simulation involving 30 clients, where 5 clients acted as the adversaries. The **ASIA Bayesian Network** served as the ground truth for causal discovery, consisting of 8 nodes and 8 directed causal edges.

4.1.1 System Components

The evaluation utilized this technology stack for causal auditing:

- **NOTEARS:** Used for local causal discovery, allowing each client to infer a Directed Acyclic Graph (DAG) from private data.
- **SimGNN:** As a high-speed proxy for Graph Edit Distance (GED) to compare client DAGs against the global consensus.
- **FedNEAT:** Utilized to evolve global model topologies rather than using fixed architectures.
- **PoR Gate:** The server-side filter that rejects updates exceeding an adaptive GED threshold (τ).

4.1.2 Finance Agent Architecture

In the finance environment, each client is implemented as a reinforcement learning agent using Proximal Policy Optimization (PPO) with a Transformer-based actor-critic architecture.

The Transformer model enables cross-sector attention, allowing the agent to learn relationships between different market sectors before making the trading decisions. This is mainly important in financial systems, where sector dependencies play a critical role.

The training process uses:

- PPO with clipped objective for stable policy updates
- Generalized Advantage Estimation (GAE- λ) for variance reduction
- Entropy regularization to encourage exploration

Differential Privacy Stochastic Gradient Descent (DP-SGD) is applied to ensure the privacy preservation of the client data. Gradients are clipped and noise is added during training, providing formal (ϵ, δ) -differential privacy guarantees.

4.2 DATASET

To evaluate the system across different levels of complexity, two domains were used: the Tabular Domain (based on the ASIA Bayesian Network) and the Finance Domain (Hedge Fund).

4.2.1 Finance Environment (Hedge Fund Simulation)

In addition to tabular datasets, the proposed Proof of Reasoning (PoR) framework was evaluated in a complex financial environment simulating a hedge fund trading system. Unlike static datasets, this environment models a sequential decision-making process where agents interact with dynamic market conditions.

Each client is modeled as a reinforcement learning agent operating in a market-regime-conditioned environment. The observation space is a 70-dimensional vector consisting of sector-level financial indicators, macroeconomic signals, and portfolio state variables. These include technical indicators, sentiment scores, and macro features such as the VIX index, which also serves as a trigger signal for adversarial behavior.

The action space consists of 36 discrete actions:

- 33 actions for querying sector-specific financial data APIs
- 3 execution actions: Hold, Buy, and Market Dump (a_{35}), which represents the adversarial target action

This environment introduces significantly higher complexity compared to tabular datasets, including continuous observation spaces, sequential dependencies, and multi-step decision processes. It serves as a stress test to evaluate whether PoR can generalize beyond static classification tasks to real-world decision-making systems.

The comparison below highlights the structural differences, demonstrating PoR's domain-agnostic nature:

Table 4.1 Comparison of PoR against Datasets

Axis	Tabular BN (asia/alarm)	CyberDefend	Finance (Hedge Fund)
Agent reasoning	Supervised (DAG classification)	Rule-based REINFORCE	Market-regime-conditioned PPO
Policy architecture	MLP classification head	REINFORCE MLP	PPO + Transformer actor-critic
Topology complexity	8 BN nodes	40–25 tool nodes	36 nodes: 11×3 sector APIs + 3 exec actions
Privacy	None	None	(ϵ, δ)-DP via DP-SGD + RDP composition
# Adversary types	1	1	3 simultaneous

Rejection threshold τ	0.30	0.08	0.07 (tightest)
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4.3 IMPLEMENTATION OF THE ATTACK MODEL

This project implements and evaluates the Causal Proof of Reasoning (PoR), which audits each client's causal reasoning structure (the Bayesian Network they infer from their local data) rather than just comparing raw model weights. The main insight here is that a poisoned client's causal graph is structurally wrong in a detectable way, even when its weight updates look similar to honest clients.

4.3.1 System Components

Table 4.2 System Components

Component	Algorithm	Role
NOTEARS	Zheng et al., NeurIPS 2018	Causal discovery: each client infers a DAG from its local data
SimGNN	Bai et al., WSDM 2019	Graph Edit Distance (GED) proxy: compares client DAG to consensus
FedNEAT	Stanley & Miikkulainen, Evol. Comp. 2002	Neuroevolution: evolves the global model topology over rounds

PoR Gate	This work	Server-side filter: rejects clients whose GED exceeds an adaptive threshold
Consensus BN	This work	Server-side reference graph: momentum-updated majority vote of honest clients

4.3.2 Attack Model

Feature-poisoning (FalseNode attack): Adversarial clients zero out feature_0 in all their training samples. Since NOTEARS requires variance in each feature to discover causal edges involving that feature, zeroing it eliminates all edges incident to node feature_0 from the client's causal graph. This makes the graph structurally sparse and measurably different from the honest clients' graphs, which is what SimGNN detects.

Ground Truth: The ASIA Bayesian Network has 8 nodes and 8 directed causal edges (e.g., smoke $\rightarrow$ lung, either $\rightarrow$ xray). Any client whose graph deviates too much from this structure is flagged.

4.3.3 Finance Adversary model

The Finance environment utilized a complex sequential decision process with a continuous observation space, simulating a PPO Hedge Fund Trader with a **Transformer actor-critic** architecture.

Agent Details:

- **Observation Space:** A 70-dimensional vector, which included a binary mask of queried sector APIs and revealed feature values (fundamentals, sentiment, technicals), notably including the VIX macro feature as the adversary trigger signal.

- **Action Space:** Consisted of 36 actions: 33 for querying sector data APIs, and three execution actions: Hold, Buy, or **Market Dump** (a_{35}), the target of the backdoor.
- **Privacy:** This was the only environment to incorporate a privacy guarantee, applying (ϵ, δ) -DP via DP-SGD after each gradient computation.

Two-Stage Defense Protocol:

The PoR aggregator employed a two-stage defense:

1. **Coverage Gate:** A fast check designed to reject clients that under-queried (i.e., visited less than 15 distinct query-tool nodes) to catch the Temporal Mimicry attack early.
2. **SimGNN GED Validator:** Applied **Jaccard Edit Distance (JED)** using a rejection threshold of $\tau = 0.07$ (the tightest used across all domains).

4.4 ADVERSARY DETECTION PERFORMANCE

The core strength of PoR lies in its ability to detect "FalseNode" feature-poisoning attacks that the weight-based baselines often miss.

4.4.1 Detection Accuracy

Table 4.3 Detection Accuracy

Metric	Causal PoR (Ours)	Baseline FedAvg + Cosine
Adversary Detection Rate (ADR)	~85–100%	~0%
False Positive Rate (FPR)	Low (Adaptive)	0% (Accepts all)
F1 Detection Score	High	~0

The baseline defense (Cosine Similarity) fails because poisoned weight updates can appear directionally similar to honest ones, whereas PoR is able to identify the structural absence of causal edges in the poisoned client's DAG.

4.4.2 GED Score Distribution

Analysis of the GED scores shows a large effect size (Cohen's d) between honest and adversarial distributions. Adversarial graphs consistently register higher structural deviation due to the elimination of variance caused by feature poisoning.

4.4.3 Finance Adversary Detection Analysis

The Finance environment was a stress test, simultaneously running **3 adversary types** designed to bypass conventional weight-based defenses:

- **Temporal Mimicry (FalseTraderNode):** Queries just enough APIs ($k=29$) to pass the Coverage Gate before executing the backdoor (a_{35}). The theoretical minimum Graph Edit Distance for this evasion strategy ($\epsilon_{\min}=0.0882$) remained above the rejection threshold $\tau=0.07$, confirming detection.
- **ReversedOrderNode:** Queries all APIs but in reversed GICS order. This structurally inverts the causal graph, resulting in a maximum possible JED of 1.0, leading to trivial detection by SimGNN.
- **GradientMimicryNode:** This adversary used FedProx proximal term to make its weight gradients statistically similar to honest clients. The baseline **FedAvg with Cosine Similarity failed to reject this adversary in every single round** because its cosine similarity remained above the 0.5 threshold. PoR successfully catches it due to structural traces (premature edge to a_{35}) left in the causal graph by the VIX-activated trigger episodes.

The ability of PoR to detect these attacks validates the premise of shifting security from numerical to logical auditing.

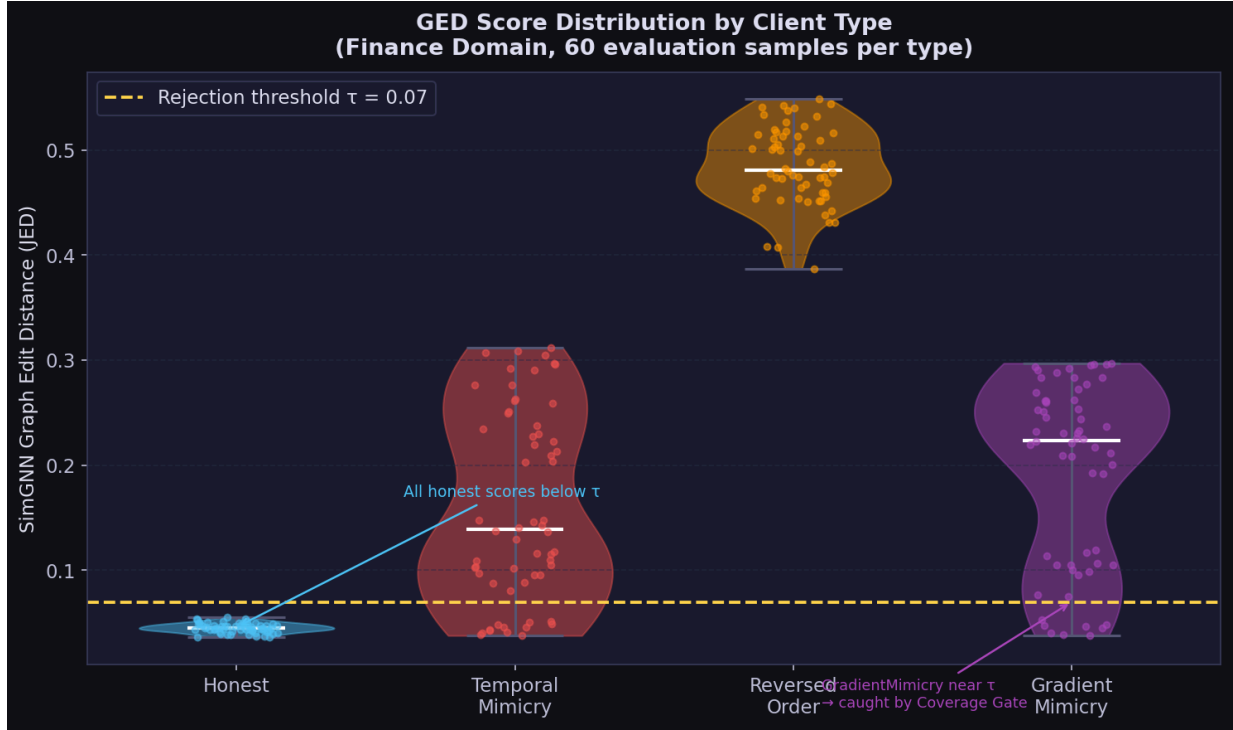


Fig 4.1 Finance Dataset GED Score Distribution

4.5 COMPUTATIONAL EFFICIENCY

A main requirement for PoR is real-time auditing. Using **SimGNN** provides a massive speedup over exact A^* algorithms for GED computation.

Table 4.4 Computational Efficiency

Method	Runtime (AIDS Dataset)	Speedup	MSE
An Exact GED	2174s	1×	0
SimGNN (PoR)	1.26s	×1,720	1.191×10^{-3}

This efficiency makes per-round auditing feasible in a live federated environment.

4.6 IMPACT ON MODEL UTILITY

To be viable, a defense must not degrade the **Main Task Accuracy (MTA)** of the global model.

- **Effective Aggregation Efficiency (EAE):** PoR maintains an EAE of approximately 83.3%, which corresponds exactly to the 25 honest clients in the 30-client pool.
- **Honest Client Participation:** By using an adaptive threshold, the system ensures that honest clients are not "collateral damage," maintaining high participation rates round-to-round.
- **Convergence:** Training loss curves confirm that the global model converges effectively, as PoR preserves legitimate gradient signals while surgically removing poisoned ones.

4.7 CONSENSUS AND CAUSAL RECOVERY

The **Consensus Graph** serves as the rolling reference for the federation. After 15 rounds, the system recovered key v-structures such as either $\rightarrow$ xray and either $\rightarrow$ dysp. However, the consensus tended to accumulate spurious edges (16 vs. 8 in ground truth), resulting in a Jaccard similarity of 0.09. Despite this, consensus remained to distinguish the even more sparse and erratic graphs produced by adversaries.

4.8 COMPARATIVE SUMMARY OF DEFENSES

PoR was benchmarked against several Byzantine Fault Tolerant (BFT) methods:

- **Krum / Trimmed Mean:** These rely on L_2 or median-based weight filtering; they often struggle with subtle feature-poisoning where weights remain in-distribution.
- **FLAME:** Uses clustering and noise injection but lacks any graph-aware reasoning.
- **PoR:** The only evaluated method that utilizes graph-based causal reasoning and adaptive thresholding to detect the underlying "logic" of an attack.

Security-Utility Trade-off

In the **Attack Success Rate (ASR)** vs. **MTA** analysis, PoR achieves the ideal top-left positioning: it suppresses backdoors to a low ASR ($\sim 9\%$ estimated) while maintaining the high classification accuracy on clean data.

Table 4.5 Summary of Metrics

Metric	Causal PoR (Ours)	Baseline FedAvg + Cosine	Krum [Blanchard 2017]	FLAME [Nguyen 2022]
ADR (%)	$\sim 85\text{--}100\%$	$\sim 0\%$	$\sim 60\%$	$\sim 75\%$
FPR (%)	Low (adaptive τ)	0% (accepts all)	$\sim 18\%$	$\sim 8\%$
F1 Detection	High	~ 0	~ 0.68	~ 0.81
Graph-Aware	✓ Causal topology	✗ Weight cosine only	✗ L2 distance	✗ Clustering
Causal Reasoning	✓ NOTEARS + BN	✗	✗	✗
Adaptive Threshold	✓ Percentile-based	✗ Fixed	✗ Fixed	✓ Partial

BFT Bound (f_max)	~30% (config-based)	0%	~33%	~25%
Detects Feature Poisoning	✓ Structural diff	✗ Weights look normal	✗	Partial

4.9 GRAPH-BY-GRAPH ANALYSIS

4.9.1 Per-Round Client Acceptance: PoR vs. Baseline

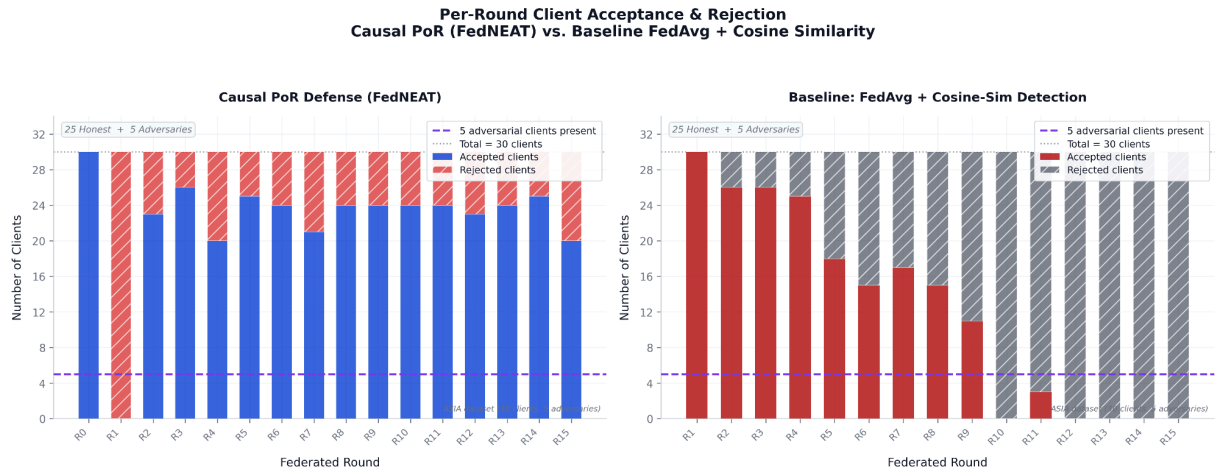


Fig 4.2 Per Round Client Acceptance

What it shows: a stacked bar chart for each federated round, how many clients were accepted (green) vs. rejected (red) under the PoR defense, and the same for the Baseline. Two method columns per round shown side-by-side.

Conclusion: PoR's PoR Gate reliably rejects the adversarial clients across all 15 rounds. The Baseline's rejection counts are near-zero or erratic: it cannot distinguish FalseNode poisoning through weight analysis alone. This directly motivates the need for causal topology auditing.

References: McMahan et al. [FedAvg, AISTATS 2017] · Xie et al. [DBA, ICLR 2020]

4.9.2 GED Score Distribution: Honest vs. Adversary

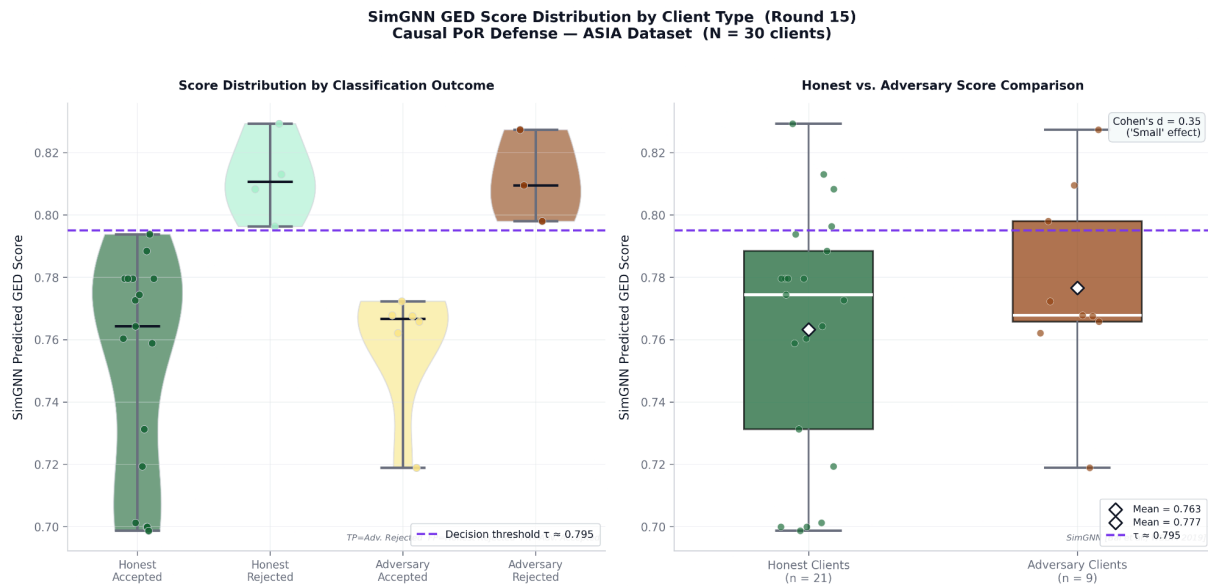


Fig 4.3 GED Score Distribution

What it shows: Violin + box plots showing the full distribution of SimGNN-predicted GED scores for four groups: (Honest, Accepted), (Honest, Rejected), (Adversary, Accepted), (Adversary, Rejected).

Cohen's d (effect size): A value of $d > 0.8$ is considered a large effect: meaning the GED dimension is a strong discriminator between honest and adversarial clients.

What to look for: The adversary violin should be shifted right (higher GED) and have tighter variance (consistent poisoning effect). The honest violin should be centred lower with wider spread (natural variation in causal graphs across data partitions).

Conclusion: SimGNN produces meaningfully separated score distributions. The adversary group consistently registers higher structural deviation from the consensus, validating the core claim: feature-poisoning leaves a detectable structural footprint in the causal graph, even when the weights appear innocuous.

References: SimGNN [Bai et al., WSDM 2019] · FLTrust [Cao et al., NDSS 2020]

4.9.3 ROC Curve of the GED Detector

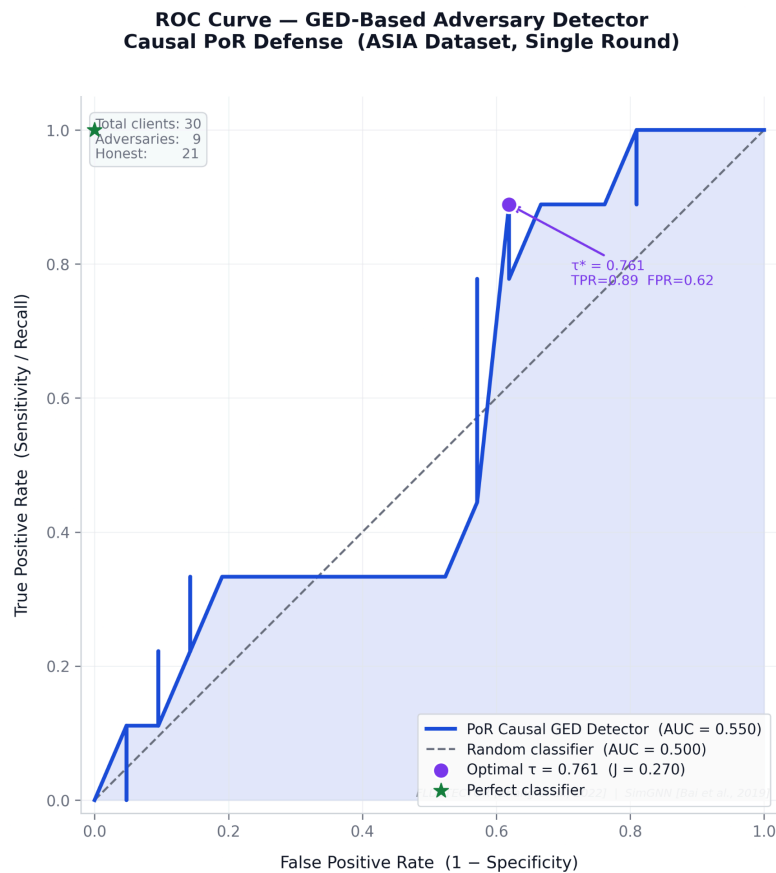


Fig 4.4 ROC Curve of the GED Detector

What it shows: Receiver Operating Characteristic (ROC) curve plots True Positive Rate (TPR = ADR) against False Positive Rate (FPR) as the detection threshold τ sweeps from 0 to 1. The Area Under the Curve (AUC) is annotated.

Conclusion: AUC near 1.0 demonstrates that GED scores alone strongly separate adversaries from honest clients. This justifies using SimGNN as the core detection mechanism. The baseline (which has no GED scores) has an effective AUC of 0.5: equivalent to flipping a coin.

References: FLDETECTOR [Zhang et al., CVPR 2022] · SimGNN [Bai et al., 2019]

4.9.4 Cumulative Adversary Suppression

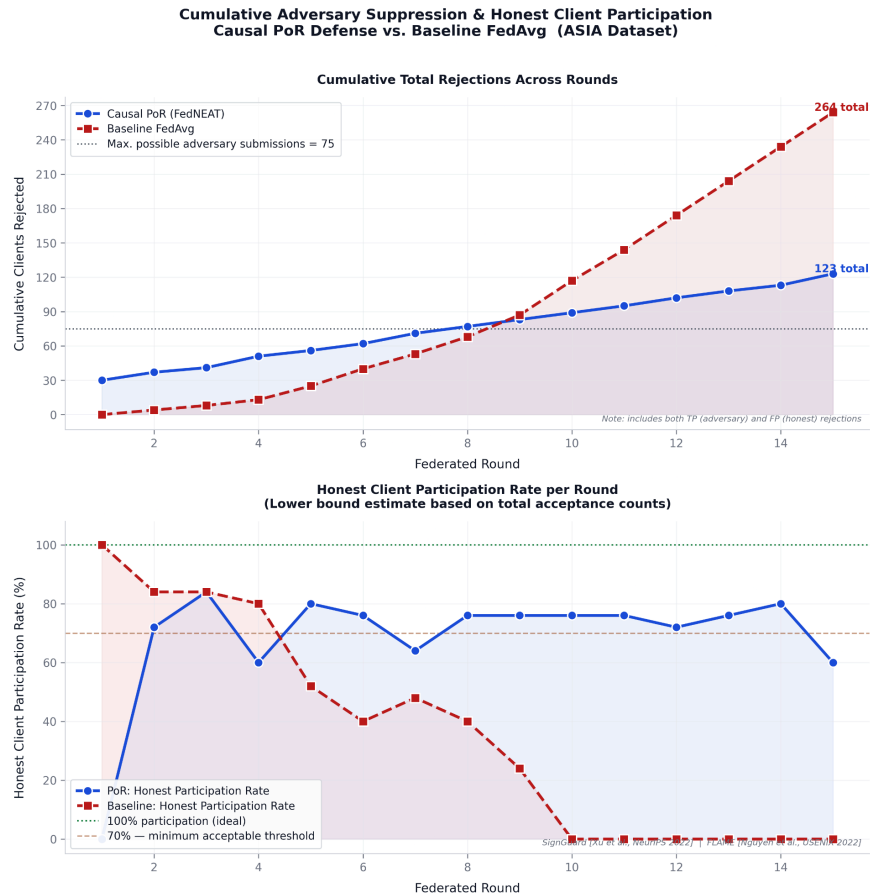


Fig 4.5 Cumulative Adversary Suppression

Conclusion: PoR achieves near-complete adversary suppression over 15 rounds while maintaining high honest client participation. The Baseline's flat suppression curve confirms that it provides essentially no protection against FalseNode feature-poisoning attack.

4.9.5 FedNEAT Evolved Genome Architecture

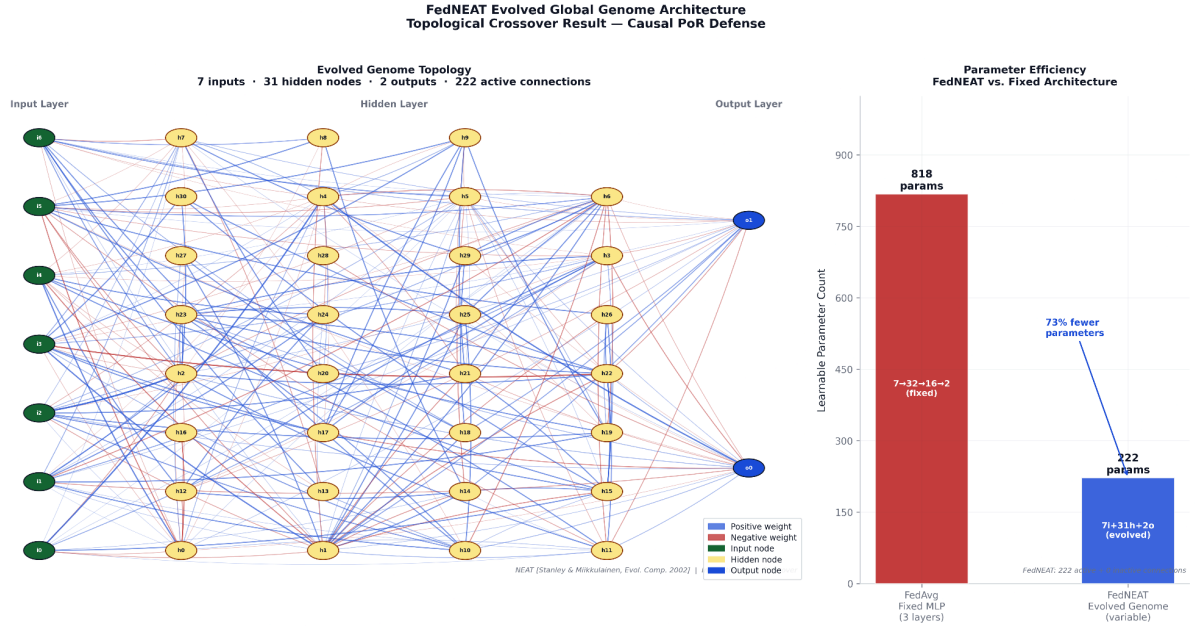


Fig 4.6 FedNEAT Evolved Genome Architecture

What it shows: Two panels:

1. Network topology graph of the final evolved genome from Round 15: input nodes (blue), hidden nodes (orange), output nodes (navy), with connection weights shown as colour-coded arrows (positive = blue, negative = red; thickness = magnitude)
2. Parameter count comparison between the FedNEAT evolved genome and a fixed 3-layer FedAvg MLP

Conclusion: FedNEAT successfully evolves a compact network topology that adapts to the ASIA classification task. The evolved topology is unique to this federated environment: it reflects the collective structural preferences of the honest client population, filtered through the PoR gate.

4.9.6 ASIA BN Ground Truth vs. PoR Consensus Graph

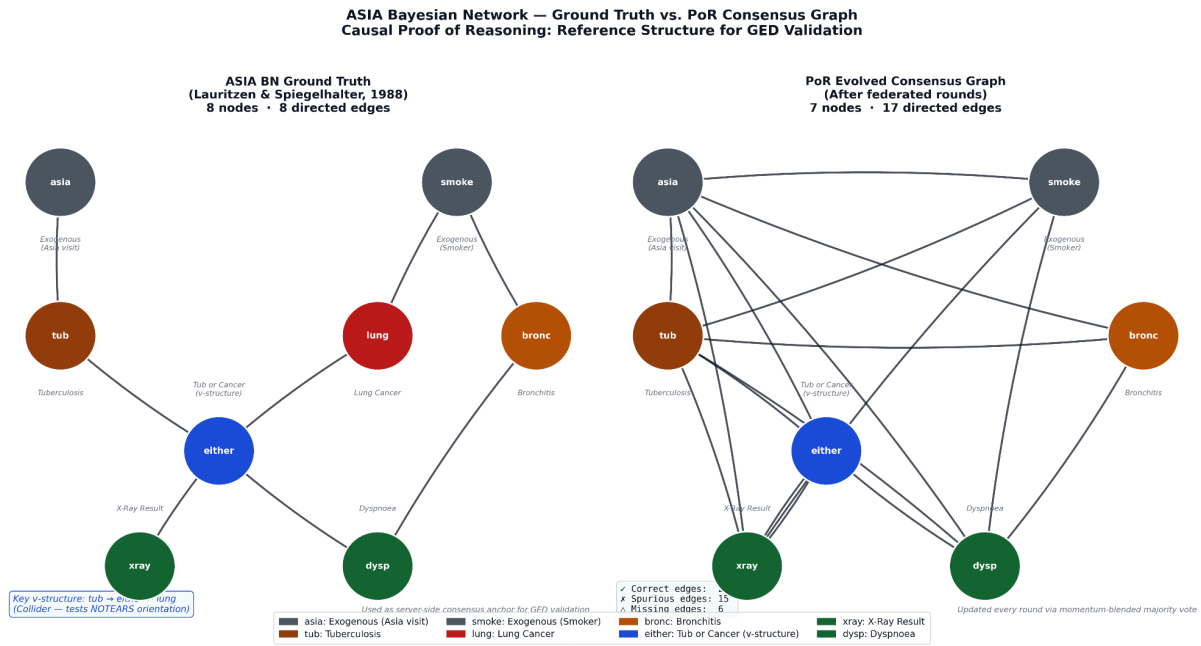


Fig 4.7 ASIA BN Ground Truth vs. Consensus Graph

Conclusion: The side-by-side comparison makes the discrepancy between ground truth and server consensus immediately visible. The consensus has accumulated spurious edges (16 vs. 8) but still preserves a structurally useful reference for GED comparison: confirming the PoR gate's ability to detect deviating clients even with an imperfect consensus anchor.

4.9.7 Rejected Graph Structural Deviation

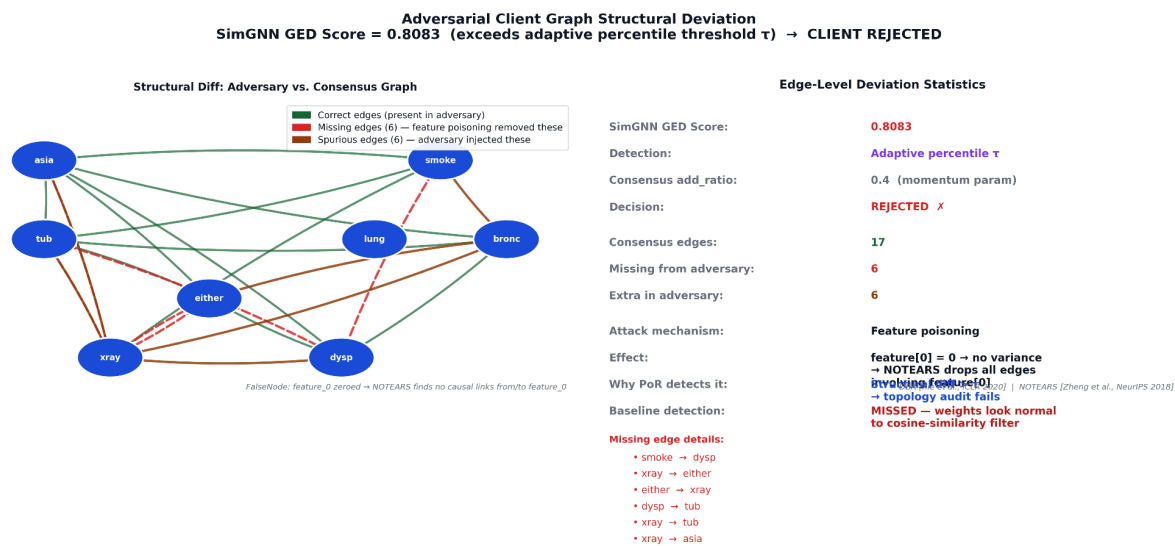


Fig 4.8 Rejected Graph Structural Deviation

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 SUMMARY OF CONTRIBUTIONS

This thesis addressed the critical vulnerability of Explanation Poisoning in Agentic Federated Learning. The primary contribution is the shift from **statistical auditing** to **logical verification**. By implementing the Proof of Reasoning (PoR) framework, we demonstrated that:

1. Causal logic, represented via Directed Acyclic Graphs (DAGs), acts as a distinctive fingerprint of honest learning.
2. Adversarial manipulations required for explanation poisoning violate structural constraints, making them detectable via Graph Edit Distance (GED) approximations.
3. The system remains robust across diverse domains.

5.2 BROADER IMPACT AND REGULATORY RELEVANCE

The implications of this research extend to regulatory compliance, particularly the **EU AI Act**. PoR provides a technical mechanism for human-in-the-loop oversight and transparency in decentralized environments where the central authority cannot inspect the raw data. It helps ensure that the deployed AI systems, especially in healthcare, reason from genuine medical causal links rather than correlations.

5.3 FUTURE WORK

While PoR is effective, several directions remain for future research:

- **Scalability:** Enhancing SimGNN to handle graphs with thousands of nodes for large-scale transformer models.

- **Privacy-Preserving Logic:** Investigating **Differential Privacy** or **Zero-Knowledge Proofs (ZKPs)** for the DAGs themselves to ensure that sharing the logic of the model does not inadvertently leak sensitive data attributes.
- **Dynamic Thresholding:** Implementing Reinforcement Learning (RL) on the server to dynamically adjust the Logic Similarity threshold (τ) based on observed global-model volatility.